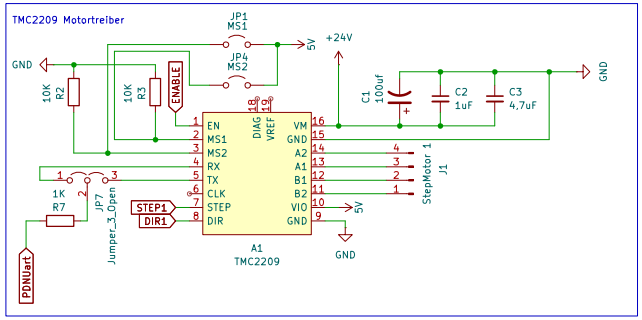
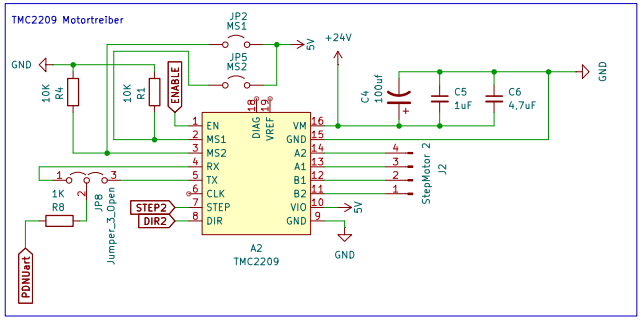
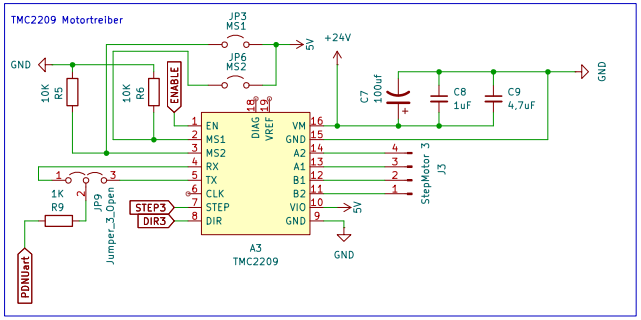
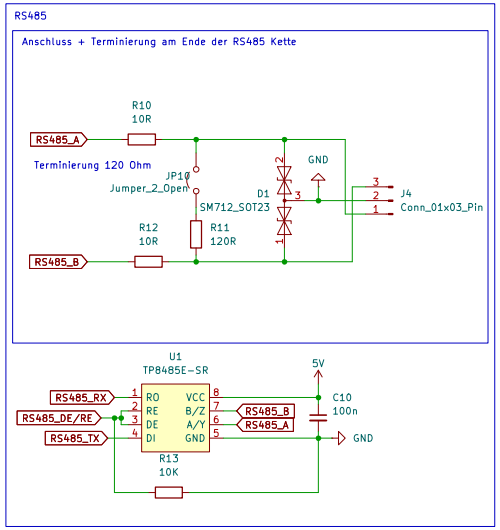
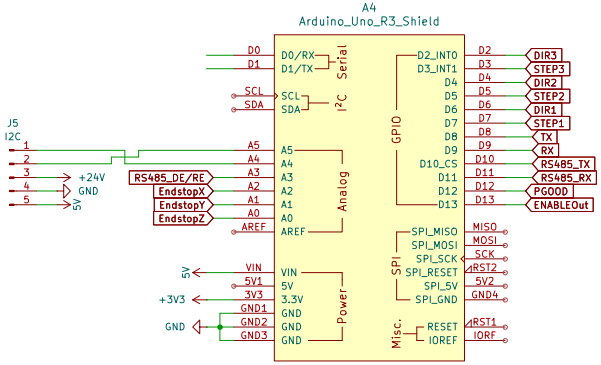
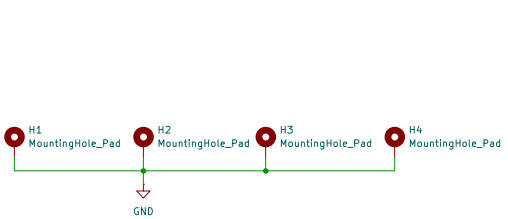

TMC2209 Motortreiber
JP1
MS1
JP4
MS2
5V
+24V
GND
10K
R2
10K
R3
ENABLE
C1
100uF
C2
1uF
C3
4,7uF
EN
MS1
MS2
RX
TX
CLK
STEP
DIR
DIAG
VREF
VM
GND
A2
A1
B1
B2
VIO
1K
R7
JP7
Jumper_3_Open
STEP1
DIR1
PDNUart
A1
TMC2209
StepMotor 1
J1

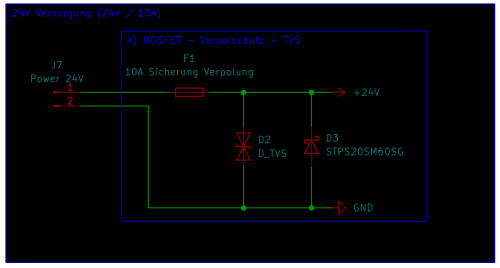

TMC2209 Motortreiber
JP2
MS1
JP5
MS2
5V
+24V
GND
10K
R4
10K
R1
ENABLE
C4
100uF
C5
1uF
C6
4,7uF
EN
MS1
MS2
RX
TX
CLK
STEP
DIR
DIAG
VREF
VM
GND
A2
A1
B1
B2
VIO
1K
R8
JP8
Jumper_3_Open
STEP2
DIR2
PDNUart
A2
TMC2209
StepMotor 2
J2

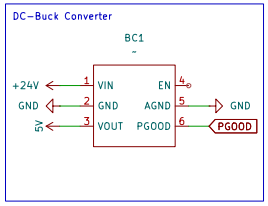

TMC2209 Motortreiber
JP3
MS1
JP6
MS2
5V
+24V
GND
10K
R5
10K
R6
ENABLE
C7
100uF
C8
1uF
C9
4,7uF
EN
MS1
MS2
RX
TX
CLK
STEP
DIR
DIAG
VREF
VM
GND
A2
A1
B1
B2
VIO
1K
R9
JP9
Jumper_3_Open
STEP3
DIR3
PDNUart
A3
TMC2209
StepMotor 3
J3

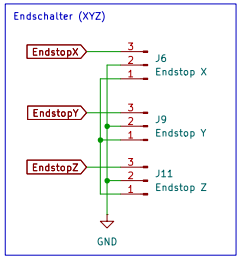

RS485
Anschluss + Terminierung am Ende der RS485 Kette
R10
10R
RS485_A
Terminierung 120 Ohm
JP10
Jumper_2_Open
GND
D1
SM712_SOT23
J4
Conn_01x03_Pin
R12
10R
R11
120R
RS485_B
U1
TP8485E-SR
5V
RS485_RX
RS485_DE/RE
RS485_TX
RO
RE
DE
DI
VCC
B/Z
A/Y
GND
RS485_B
RS485_A
C10
100n
R13
10K

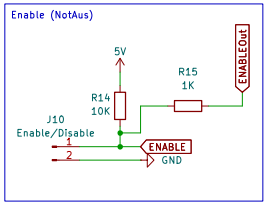

A4
Arduino_Uno_R3_Shield
J5
I2C
+24V
GND
5V
RS485_DE/RE
EndstopX
EndstopY
EndstopZ
D0
D1
SCL
SDA
A5
A4
A3
A2
A1
A0
AREF
VIN
5V
3.3V
GND
GND1
GND2
GND3
Serial
I2C
Analog
Power
D2_INT0
D3_INT1
D4
D5
D6
D7
D8
D9
D10_CS
D11
D12
D13
GPIO
DIR3
STEP3
DIR2
STEP2
DIR1
STEP1
TX
RX
RS485_TX
RS485_RX
PGOOD
ENABLEOut
SPI_MISO
SPI_MOSI
SPI_SCK
SPI_RESET
SPI_5V
SPI_GND
SPI
RESET
IOREF
Misc.
MISO
MOSI
SCK
RST2
5V2
GND4
RST1
IORF

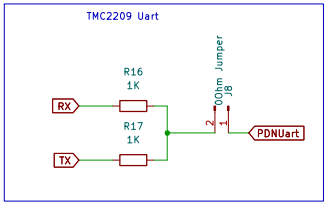

H1
MountingHole_Pad
H2
MountingHole_Pad
H3
MountingHole_Pad
H4
MountingHole_Pad
GND


24V Versorgung (24V / 10A)
A) MOSFET – Verpolschutz + TVS
J7
Power 24V
F1
10A Sicherung Verpolung
+24V
D2
D_TVS
D3
STPS20SM60SG
GND


DC-Buck Converter
BC1
+24V
GND
5V
VIN
GND
VOUT
EN
AGND
PGOOD
GND
PGOOD


Endschalter (XYZ)
EndstopX
EndstopY
EndstopZ
J6
Endstop X
J9
Endstop Y
J11
Endstop Z
GND


Enable (NotAus)
5V
R14
10K
R15
1K
ENABLEOut
J10
Enable/Disable
ENABLE
GND


TMC2209 Uart
R16
1K
R17
1K
RX
TX
00hm Jumper
J8
PDNUart